

Thach Thanh Nguyen

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Final-year Microelectronics Engineering student at HUFLIT with hands-on experience in schematic design, PCB layout/routing, board bring-up, debugging, and board testing with STM32, ESP32, and Raspberry Pi.

EDUCATION

Ho Chi Minh City University of Foreign Languages – Information Technology (HUFLIT)

2023 – Present

B.Sc. Information Technology | Major: Microelectronics Engineering

AWARDS

- Top 12 — Green Innovation Contest, HUST, 2025
- Top 20 — BK Innovation, HCMUT, 2025
- University-level research project, 2025

PROJECTS

Buck vs LDO Energy Comparison

01/2026 – 08/2026

- Designed and brought up a 4-layer KiCad PCB with jumper-selectable TPS62840 buck / STLQ020 LDO branches powering an STM32L432 + BME280 node, and wrote the bare-metal firmware for the 5 s BME280–Stop2 duty cycle (race-to-sleep).
- Built an INA228 input-side energy rig (approximately 10k samples/s) and ran 286 verified experiments across 3.30–4.20 V — each run as a statistical unit, with confidence intervals plus ABBA and marker verification campaigns.

GitHub: github.com/NgThachThanh/stm32-buck-vs-ldo

GreenEco — Smart Greenhouse

2025

- Raspberry Pi greenhouse system: a REST API driving 4 active-low relay channels (pump, two fans, grow light), with sensor data streaming to the team's server and control app.
- Brought up the hardware and sensor layer on two buses — RS485/Modbus RTU (ES-Soil7 soil, DFRobot environmental sensor) and UART (SEN0220 CO₂, SEN0501) — plus relay GPIO mapping, with Python unit tests covering the GPIO, UART/Modbus and API layers.

GitHub: github.com/NgThachThanh/greeneco

Smart Helmet

2025

- ESP32-S3 dual-core FreeRTOS firmware: 100 Hz MPU6050 sensing task with a filtered, speed/road-adaptive threshold fall-detection state machine (3.5 g final threshold, staged alerts, physical cancel button).
- Automatic SMS alert via GSM module (single send per crash, 10 s cancel window) and an OLED HUD fed by BLE navigation data from the phone.

GitHub: github.com/NgThachThanh/smart-helmet

PUBLICATIONS

Efficient FPGA-Based Hardware Noise Filtering and Real-Time Motion Detection Using RE200B Sensor on Trion T8 BGA81

Springer CCIS, 2026 | Published

Research focus: FPGA-based hardware noise filtering and real-time motion detection.

Output: Conference paper | *Co-author* — Computational Intelligence in Engineering Science

DOI: https://doi.org/10.1007/978-3-031-98167-8_21

Toward a Hybrid Efficient Edge-Host Smart Helmet System for Drowsiness Detection and Accident Verification using ESP32-S3

CUTE 2026 | Accepted

Research focus: Edge-host smart-helmet system for drowsiness detection and accident verification.

Output: Conference paper | *Co-author* — The 5th International Conference on Sustainable Development in Civil, Urban and Transportation Engineering

DG-GhostESP-96: dark-guided low-light image enhancement on STM32H750VBT6

ICEBA 2026 | Accepted

Research focus: Dark-guided low-light image enhancement on STM32H750.

Output: Conference paper | *Co-author* — The 7th International Conference on Engineering, Physics, MEMS-Biosensors and Applications

TECHNICAL SKILLS

PCB workflow: Schematic, layout/routing, board bring-up, debugging, validation

Programming: C/C++, Python, Verilog

MCU: STM32, ESP32

Protocols: SPI, I2C, UART, RS485 (Modbus RTU), BLE

Tools: KiCad, Raspberry Pi, Git, GitHub, Linux

LANGUAGES

Vietnamese (native) | English (intermediate)